



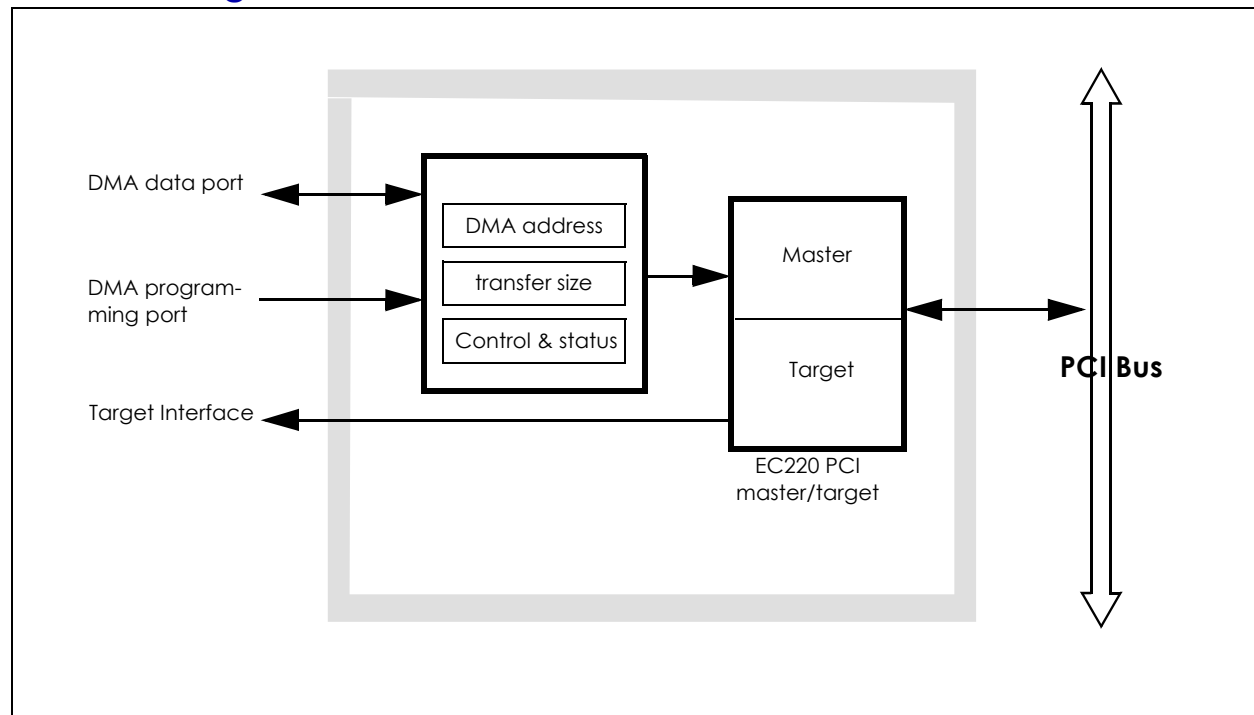
## 1 Introduction

This application note describes optional DMA function for the EC220 or similar PCI master/target or bridge cores. The standard PCI master/target core allows the user to access the PCI bus directly by specifying request by using the user interface signals. This method provides fast access to the PCI bus in real time. The DMA options allows the user to program the DMA registers and then lets the DMA controller to take over the transfer. This method in general requires less interaction between the PCI core and the user logic.

The DMA option does not affect the target operation of the PCI master/target core. The target function, however, can be used to program the DMA controller register if desire. This will be explained in later sections.

The following diagram illustrates the DMA and PCI master/target functions.

## 2 Block Diagram





## 3 Programming Model

The DMA function is controlled by software which program several control registers inside the DMA controller. The following table illustrates the control registers.

**Table 1: DMA Control Registers**

| Offset                 | Bits  | Descriptions                                                                                                                                                                                         |
|------------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DMA address register   |       |                                                                                                                                                                                                      |
| 00                     | 31:2  | Starting PCI address                                                                                                                                                                                 |
|                        | 1:0   | These two bits are hardwired to zero because DMA operates at size of 32-bit words                                                                                                                    |
| Transfer size register |       |                                                                                                                                                                                                      |
| 01                     | 31:21 | Hardwired to zero                                                                                                                                                                                    |
|                        | 20:0  | The number of 32-bit words to be transferred. The maximum is 2Meg-1.                                                                                                                                 |
| DMA status register    |       |                                                                                                                                                                                                      |
| 10                     | 31:4  | Hardwired to zero                                                                                                                                                                                    |
|                        | 3     | DMA direction. If this bit is “0”, it indicates DMA transfer data from PCI to user port. If it is “1”, data is transferred from user port to PCI.                                                    |
|                        | 2     | Start Bit. DMA starts when user writes a “1” to this bit. This bit is reset to “0” when DMA finishes                                                                                                 |
|                        | 1     | Interrupt bit. This bit is set to “1” when DMA completes. It can be used to trigger interrupt at the completion of DMA. This bit is cleared when user writes a “1” to it. Writing “0” has no effect. |
|                        | 0     | Error bit. This bit is set to “1” when DMA is terminated due to error on the PCI bus. This bit is cleared when user writes a “1” to it. Writing “0” has no effect.                                   |



## 4 User Interface Signals

The DMA option replaces the master side of the user interface by DMA programming port and data port. The following table lists the new signals. Signals with “#” suffix are active low signals.

**Table 2: DMA Programming Port**

| Signal       | Type | Descriptions                                                                                                                     |
|--------------|------|----------------------------------------------------------------------------------------------------------------------------------|
| CR_ADDR[1:0] | I    | DMA control register address.                                                                                                    |
| CR_ADS#      | I    | Access to DMA control registers.                                                                                                 |
| CR_BE#[3:0]  | I    | Byte enable to DMA control registers.                                                                                            |
| CR_DRD[31:0] | O    | Read data from DMA control registers.                                                                                            |
| CR_DWR[31:0] | I    | Write data to DMA control registers.                                                                                             |
| CR_RDY#      | O    | Ready signal to indicate completion of DMA control register access.                                                              |
| CR_WR        | I    | Read write indicator. If this signal is “1”, the DMA control register is written to, otherwise the DMA control register is read. |

**Table 3: DMA Data Port**

| Signal      | Type | Descriptions                                                                                                          |
|-------------|------|-----------------------------------------------------------------------------------------------------------------------|
| D_CNT[20:0] | O    | Data count. This value is incremented by 1 for each data transferred. At the first data transfer, this value is zero. |
| D_DRD[31:0] | I    | Data transferred from user port to PCI bus.                                                                           |
| D_DWR[31:0] | O    | Data transferred from PCI bus to user port.                                                                           |
| D_ERR#      | O    | This signal has the opposite value of the error bit in the status register.                                           |
| D_INTR#     | O    | This signal has the opposite value of the interrupt bit in the status register. It can be used to trigger interrupt.  |

**Table 3: DMA Data Port**

| Signal    | Type | Descriptions                                                                                                                                                                                       |
|-----------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D_RDY_RD# | O    | This signal is asserted for each data read from D_DRD to be transferred to the PCI bus. The current data on D_DRD is captured and the next data should be on the D_DRD bus in the following cycle. |
| D_RDY_WR# | O    | This signal is asserted for each data written to D_DWR which is received from the PCI bus. The user logic must capture the D_DWR data when D_RDY_WR# is asserted.                                  |
| D_START   | O    | This signal has the same value as the start bit of the status register.                                                                                                                            |
| D_WR      | O    | This signal has the same value as the direction bit of the status register.                                                                                                                        |

## 5 Accessing the DMA Registers

The CR\_ADS# signal is asserted by the user logic to indicate read or write to the control registers. CR\_WR must be high to signal register write. If CR\_WR is low when CR\_ADS# is asserted, it is a read access. CR\_WR will remain valid until the end of read or write. The address of the control register is presented at the same time and remains valid until the write is completed. Write data must be valid one cycle after CR\_ADS# and remains valid until the end of the transfer. Write completion is indicated by the CR\_RDY# signal asserted by the control register.



When CR\_RDY# is asserted, write to the control register is completed.

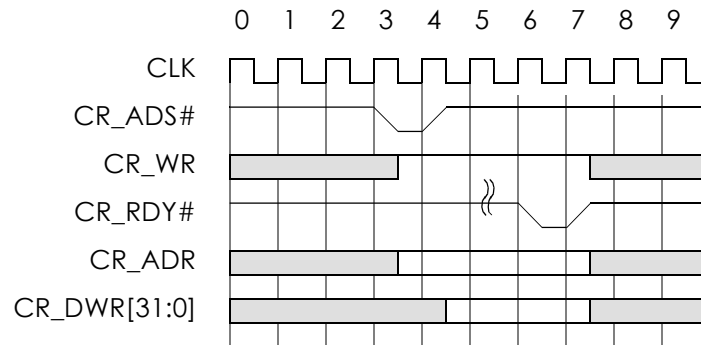


Figure 5.1 Control register write

Similar timing sequence is used for control register read except that CR\_WR signal is low.

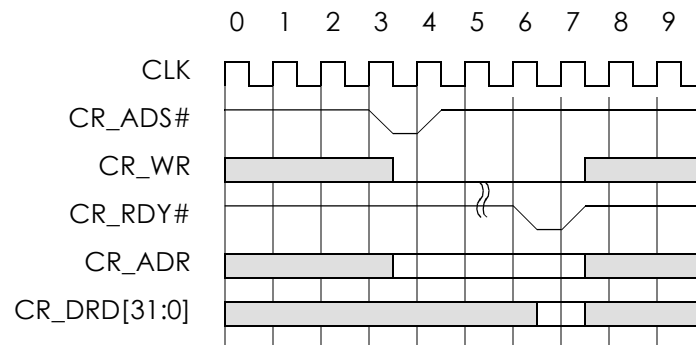


Figure 5.2 Control register read



## 6 DMA Data Transfer

To program a DMA transfer, the user should first program the DMA address and transfer registers. The DMA can then be started by writing a “1” to the start bit in the status register with the proper direction bit.

Once DMA is started, the PCI bus master would start to read or write data from the PCI bus and transfer data to the DMA data port. The PCI bus master would use burst transfer as long as possible to transfer all the DMA data. It is expected the length of the transfer to be limited by the latency timer in the PCI configuration register.

The following diagram illustrates DMA transfer of five 32-bit words from PCI to DMA data port. Please note that the timing of the D\_RDY\_WR# signal controlled by the speed of the PCI bus. It may be asserted with idle cycles (wait states) in between data or it can be asserted multiple cycles in a row for consecutive data.

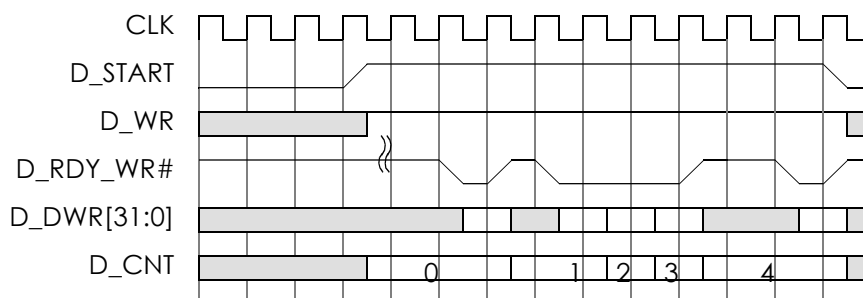


Figure 6.1 DMA from PCI to user port

The following diagram illustrates DMA transfer of five 32-bit words from user port to PCI. Please note that the timing of the D\_RDY\_RD# signal controlled by the speed of the PCI bus. It may be asserted with idle cycles (wait states) in between data or it can be asserted multiple cycles in a row for consecutive data. The first data must be presented at the D\_DRD bus when the start bit is asserted and remains valid until D\_RDY\_RD# is asserted. Except for the last data of the transfer, the next data must be valid at the D\_DRD bus one cycle after the asser-



tion of the D\_RDY\_RD# signal.

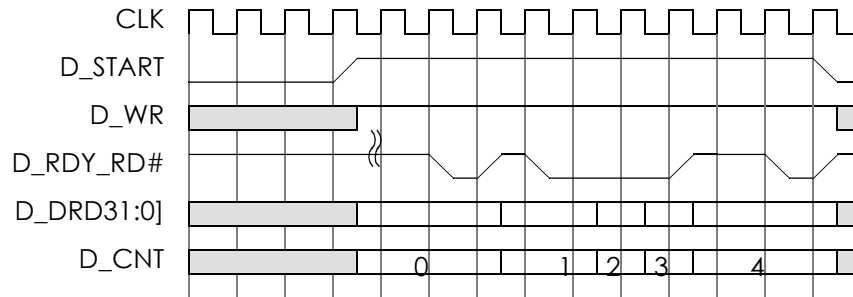


Figure 6.2 DMA from user port to PCI

## 7 Program the DMA from the PCI Bus

It is possible to program the DMA from the PCI bus by making use of the PCI target port. This can be achieved without changing the core at all. The following signals can be connected together to give control of the DMA programming port to the PCI bus.

**Table 4: DMA Programming Port**

| DMA programing port | Target side user interface |
|---------------------|----------------------------|
| CR_ADDR[1:0]        | H_AOUT[3:2]                |
| CR_ADS#             | H_ADSM#                    |
| CR_BE#[3:0]         | H_BEOUT#[3:0]              |
| CR_DRD[31:0]        | H_DIN[3:0]                 |
| CR_DWR[31:0]        | H_DOUT[3:0]                |
| CR_RDY#             | H_RDYH#                    |
| CR_WR               | H_WROUT                    |
| "1"                 | H_ABORT#                   |



**Table 4: DMA Programming Port**

| DMA programing port | Target side user interface |
|---------------------|----------------------------|
| "0"                 | H_BII#                     |
| Open                | H_BLAST#                   |
| Open                | H_BOFF#                    |
| Open                | H_CS#                      |
| "1"                 | H_RETRY#                   |
| D_INTR#             | H_INTA# *                  |

\* D\_INTR# and H\_INTA# are optional connection. This would trigger interrupt on the PCI bus when DMA completes.

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